

# Quiz 9

Info ~~veg~~  $\equiv$   $\equiv$

N  $\equiv$  Y  $\equiv$  N Y Y  
6 7 8 9 13

$$\text{Info}_{\equiv}(D) = \frac{1}{5} I(1,0) + \frac{4}{5} I(3,1)$$

$$= \frac{1}{5} \left( -\frac{1}{5} \log_2 \left( \frac{1}{5} \right) - \frac{0}{5} \log_2 \left( \frac{0}{5} \right) \right) + \frac{4}{5} \left( -\frac{4}{5} \log_2 \left( \frac{4}{5} \right) - \frac{1}{5} \log_2 \left( \frac{1}{5} \right) \right)$$

$$= 0.028 + (-0.114)$$

$$= -0.146$$

$$\text{Info}_{\equiv}(D) = \frac{2}{5} I(1,1) + \frac{3}{5} I(2,1)$$

$$= \frac{2}{5} \left( -\frac{1}{5} \log_2 \left( \frac{1}{5} \right) - \frac{1}{5} \log_2 \left( \frac{1}{5} \right) \right) + \frac{3}{5} \left( -\frac{2}{5} \log_2 \left( \frac{2}{5} \right) - \frac{1}{5} \log_2 \left( \frac{1}{5} \right) \right)$$

$$= 0 +$$